

MATLAB MCP Server Setup Guide

For Bridges Using a TOML or JSON Configuration File

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Overview

This guide does not assume which local model bridge is installed on the target machine, and does not name any specific product. Instead it splits by the **shape of that bridge's own configuration file**, since that alone is usually enough to tell which method applies:

- **Method A** — for bridges that store configuration as **TOML** (recognizable by a `config.toml`-style file with section headers written like `[[mcp_servers]]`).
- **Method B** — for bridges that store configuration as **JSON** (recognizable by curly braces and a key such as `"mcpServers":\{...\}`), or for any bridge whose format is unknown.

Open whatever configuration file the bridge already has to see which shape it is, then follow the matching method within each step below. **Try Method A first if unsure; if the config file does not look like TOML, skip straight to Method B.**

★ Note

Steps that do not depend on the bridge's configuration format (downloading the files, raising the context limit) are given only once, since the two methods would be identical anyway.

What you will need:

- A Windows PC
- A stable internet connection, for the downloads in Step 1 only
- Enough free disk space for the MCP server binary and the skill catalogs (well under 1 GB)

Two assumptions this guide relies on:

Assumption 1 — A Bridge Already Exists

The target machine already has a working local model connector bridge connecting a local LLM to MCP servers. Tool-calling infrastructure is a given — there is no need to test whether the model itself has file or tool access before registering anything.

Assumption 2 — Context Window Is Not a Hard Constraint

The context limit of the local model can be increased as needed, so the full MATLAB/Simulink tool set and skill content can be loaded in full, without pruning or selective loading.

Step 1. Download Everything Into One Folder

This step is identical either way — it does not depend on the bridge's configuration format. Everything is downloaded by hand from the browser and extracted into one working folder. By the end of this step, that folder looks like this:

```
C:\mcp-deploy\matlab\  
matlab-mcp-server\      <- MCP server binary          (Step 2 below)  
matlab-agentic-toolkit\  <- skills catalog + templates    (Step 3 below)  
simulink-agentic-toolkit\ <- Simulink tools/skills, optional (Step 4 below)
```

1. Create the working folder: C:\mcp-deploy\matlab.
2. Inside it, create a subfolder named matlab-mcp-server. Open <https://github.com/matlab/matlab-mcp-server/releases> in a browser, open the **latest** release, download the .exe listed there for your platform, and save it into that subfolder.
3. Go back to C:\mcp-deploy\matlab, then:
 - (a) Open <https://github.com/matlab/matlab-agentic-toolkit>.
 - (b) Click the green **Code** button, then choose **Download ZIP**.
 - (c) Right-click the downloaded ZIP and choose **Extract All**.
 - (d) Click **Browse**, set the destination to C:\mcp-deploy\matlab itself — do not accept whatever folder is pre-filled — then click **Extract**.
 - (e) **Rename** the extracted matlab-agentic-toolkit-main folder to matlab-agentic-toolkit.
4. Only if Simulink tools/skills are also needed, still inside C:\mcp-deploy\matlab:
 - (a) Open <https://github.com/matlab/simulink-agentic-toolkit>.
 - (b) Click **Code** → **Download ZIP**.
 - (c) Extract it the same way, setting the destination to C:\mcp-deploy\matlab itself.
 - (d) **Rename** the extracted simulink-agentic-toolkit-main folder to simulink-agentic-toolkit.

Step 2. Raise the Context Limit (Optional)

Also identical either way — there is no one universal setting name, so the same general guidance applies regardless of configuration format. Do this if the bridge and model in use allow it; skip it otherwise and come back to it later if tool calls start failing.

1. If a context-length or token-limit setting is exposed for whatever is actually running the model underneath the bridge, raise it comfortably above what a full tool set plus skill text is expected to need.

★ Note

A too-small context limit here will **silently truncate** tool definitions and skill content instead of erroring cleanly — err generous if adjustable, or shrink the tool/skill selection if not.

Step 3. Register the MCP Server

► Method A — If the Bridge Uses a TOML Config File

1. Open the bridge's TOML config file — commonly `config.toml` in a hidden dot-folder in your home directory.
2. Add this block:

```
[[mcp_servers]]
name = "matlab-mcp"
transport = "stdio"
command = "C:/mcp-deploy/matlab/matlab-mcp-server/matlab-mcp-server-windows-x64.exe"
args = ["--extension-file=C:/mcp-deploy/matlab/simulink-agentic-toolkit/tools/tools.json"]
```

► Method B — If It Uses JSON, or Anything Else

1. Locate the bridge's config file that lists MCP servers — commonly JSON, with an mcpServers object.
2. Back up the file.
3. Add this entry, or copy a starting template from matlab-agentic-toolkit/templates/:

```
{
  "mcpServers": {
    "matlab-mcp": {
      "command":
        "C:\\mcp-deploy\\matlab\\matlab-mcp-server\\matlab-mcp-server-windows-x64.exe",
      "args": [
        "--extension-file=C:\\mcp-deploy\\matlab\\simulink-agentic-toolkit\\tools\\tools.json"
      ]
    }
  }
}
```

Step 4. Place the Skills

► Method A — If the Bridge Uses a TOML Config File

1. Find the bridge's skills folder — check its documentation for the exact location, or look for a folder named .agents/skills/ (project-local) or ~/.agents/skills/ (user-global) in your home directory. Depending on the bridge, this folder may not exist yet — create it if it doesn't.
2. **Set \$dst below to that skills folder**, then run this in PowerShell — skills-catalog nests each skill inside a category folder, so this finds every SKILL.md wherever it sits and copies just that skill's folder straight into \$dst, for both toolkits in one pass:

```
$dst = "<bridge's skills folder>"
$srcs = "C:\mcp-deploy\matlab\matlab-agentic-toolkit\skills-catalog",
        "C:\mcp-deploy\matlab\simulink-agentic-toolkit\skills-catalog"
foreach ($src in $srcs) {
  if (Test-Path $src) {
    Get-ChildItem $src -Directory -Recurse | Where-Object {
      Test-Path (Join-Path $_.FullName "SKILL.md")
    } | ForEach-Object {
      Copy-Item $_.FullName -Destination (Join-Path $dst $_.Name) -Recurse -Force
    }
  }
}
```

The second toolkit path is simply skipped if it was not downloaded.

3. If the bridge's config file has an enabled_skills list, add the copied skill names to it, then reload or restart.

► **Method B — If It Uses JSON, or Anything Else**

- **If it expects skills at a fixed path:** set \$dst to that path and run the same PowerShell command shown above under Method A — it copies every skill folder from both toolkits in one pass and simply skips whichever toolkit was not downloaded.

Step 5. Restart and Verify

1. Restart or reload the bridge, whichever it supports, so the configuration change takes effect.
2. Confirm the MATLAB server appears in the bridge's tool list, with its full tool set, not a truncated one.
3. Ask your agent:

What version of MATLAB is running? List the installed toolboxes.